

Roopak Reddy

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CAREER OBJECTIVE

B.Tech CSE (Artificial Intelligence & Machine Learning) graduate (2025) with a strong foundation in data structures, algorithms, and object-oriented programming across Python, Java, C, and C++. Experienced in building scalable, event-driven backend systems (FastAPI, Kafka, Docker, AWS), automating business workflows on ERPNext/Frappe, and applying machine learning to real-world problems. Seeking a Software Development / Application Engineering role where I can design reliable, distributed systems, collaborate closely with cross-functional teams, and keep growing as an engineer.

WORK EXPERIENCE

Software Developer — ThinkCampus Pvt Ltd

August 2025 – Present

- Designed and built scalable Python microservices using FastAPI, Flask, Docker, and AWS, enabling loosely coupled, high-performance distributed applications.
- Implemented asynchronous, event-driven processing with Apache Kafka to reliably handle high-volume, distributed event streams for policy, claims, and notification workflows.
- Extended ERPNext with custom Frappe DocTypes, server scripts, and REST APIs, applying object-oriented design and Agile/SDLC practices across the development lifecycle.
- Collaborated with cross-functional teams on system design and architecture discussions, debugging production issues and delivering reliable, maintainable features.
- Documented workflows and troubleshooting guides, and participated in code reviews to improve backend performance and team self-sufficiency.

PROJECTS

SDN Controller Network Load Balancing for Cloud Data Centers (Distributed Systems)

- Designed a load-balancing approach for an SDN controller to distribute traffic across a simulated cloud data center, using HAProxy-based traffic-distribution strategies to improve availability and reduce latency.
- Applied SDN and OpenStack concepts to optimize resource allocation across distributed infrastructure.

eBao InsurTech Platform Integration

- Built Python/FastAPI microservices integrating eBao Life II insurance APIs, using Kafka for reliable, fault-tolerant event processing across policy, claims, and lapse workflows.
- Designed PostgreSQL schemas for policy and audit data, reducing report generation time by 40% through schema and query optimization.

Text Toxicity Detection (Algorithms / ML)

- Built an NLP pipeline (tokenization, POS tagging) and applied ML classification algorithms (clustering, SVM) to detect toxic content, balancing classification accuracy with computational efficiency.

Enterprise E-Commerce Platform — Frappe Framework (Software Design)

- Architected role-based access control (RBAC) with custom authorization middleware, applying multi-tier design principles to separate secure internal workflows from public-facing operations at scale.

EDUCATION

B.Tech, CSE (Artificial Intelligence & Machine Learning)

2021 – 2025

CMR College of Engineering and Technology · CGPA 7.07

B.Tech Minor, CSE (Cyber Security)

2023 – 2025

CMR College of Engineering and Technology · CGPA 8.03

Intermediate (MPC)

2019 – 2021

Sri Amogha Junior College · 82.6%

SKILLS

Programming Languages: Python, Java, JavaScript, C, C++, SQL, HTML, CSS

Core CS: Data Structures & Algorithms, Object-Oriented Programming, Complexity Analysis, Distributed Systems, REST API Design, Software Design

Frameworks/Platforms: FastAPI, Flask, Docker, AWS, Kafka, ERPNext, Frappe Framework, PostgreSQL

Data/ML: Pandas, NumPy, scikit-learn, NLP basics (tokenization, POS tagging)

Practices & Tools: Agile, SDLC, Code Reviews, Git, VS Code, Jupyter Notebook

Soft Skills: Problem-Solving, Technical Communication, Mentoring, Teamwork, Leadership, Time Management

CERTIFICATIONS

- Python Certification — CISCO
- Certified AI Associate — Salesforce
- Cybersecurity Fundamentals Associate — Zscaler
- Google Analytics Certification
- ERPNext Completion Certificate — MaxMunus
- Claude (Anthropic) Certificate

ACTIVITIES

- Organized UI & logo design competition; participated in inter-college football tournaments and athletics.
- Completed French language classes; active in team/community engagement (Street Cause).